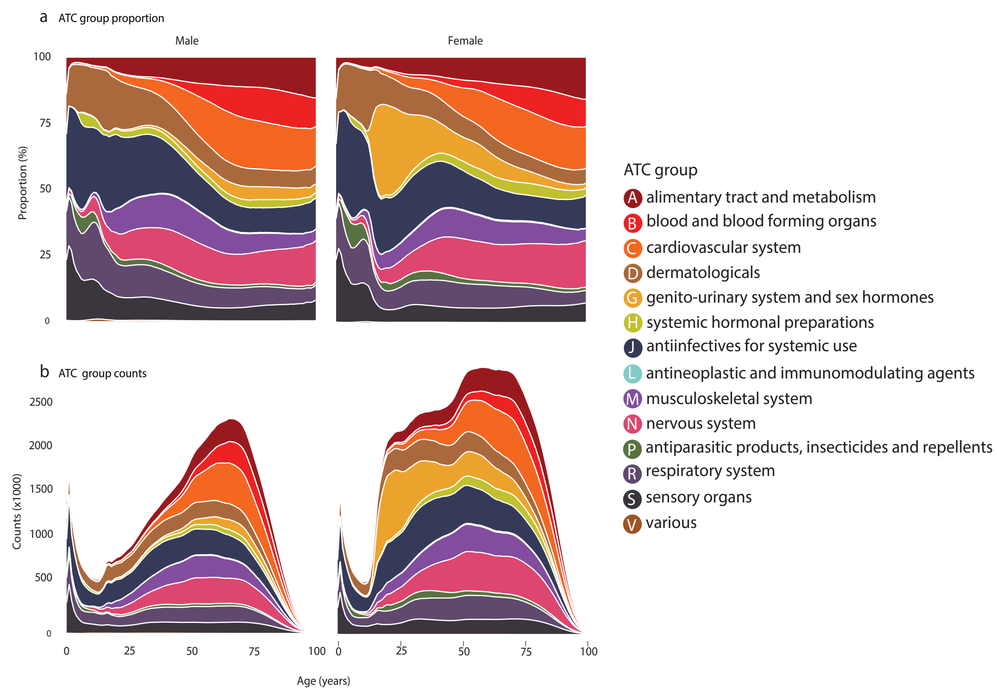
POPULATION-WIDE ANALYSIS OF PRESCRIPTION TRAJECTORIES: 7.2 MILLION DANISH PATIENTS OVER 25 YEARS WITH 1.1 BILLION REDEEMED PRESCRIPTIONS

IDEA

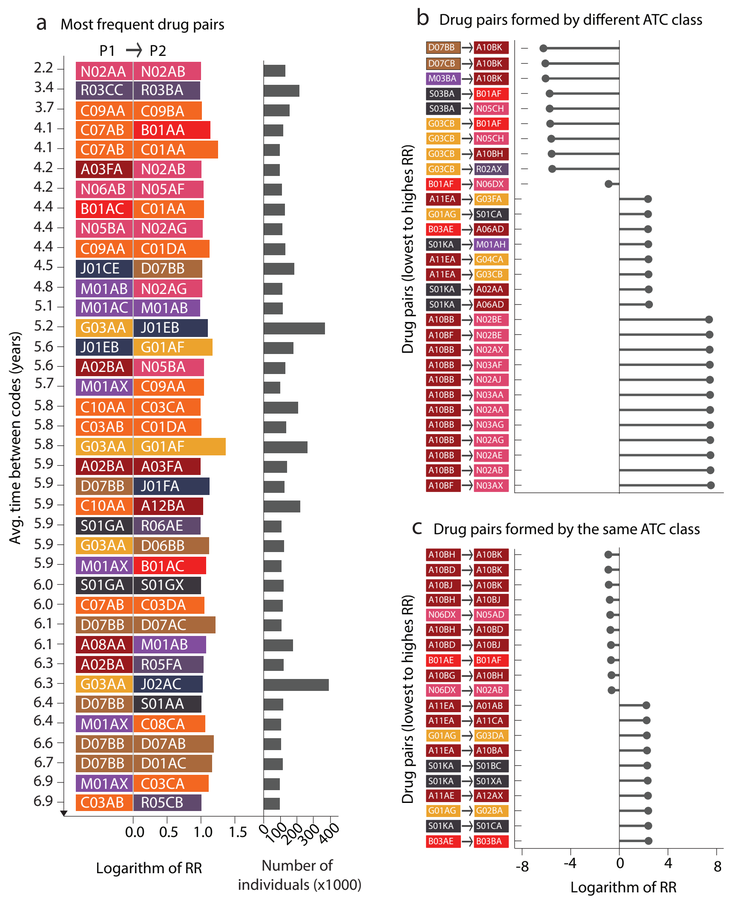
It is unknown how sequential drug patterns convey information on a patient's health status and treatment guidelines rarely account for this. Drug-agnostic longitudinal analyses of prescription trajectories in a population-wide setting are needed. Using 25 years of data from the Danish prescription registry we model the risk of sequentially redeeming drugs after each other and find the "high-ways" in drug space.

METHODS

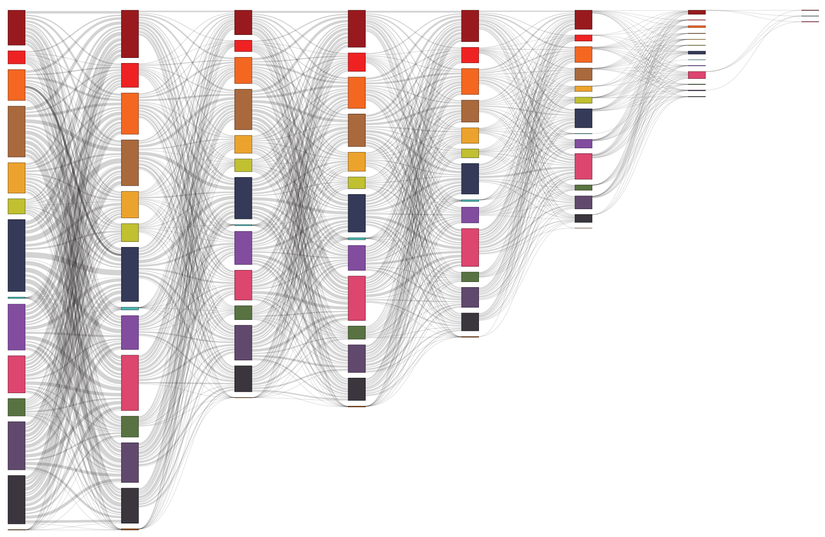
A) Retrieve data from Prescription Registry



B) Compute Relative Risks of drug pairs

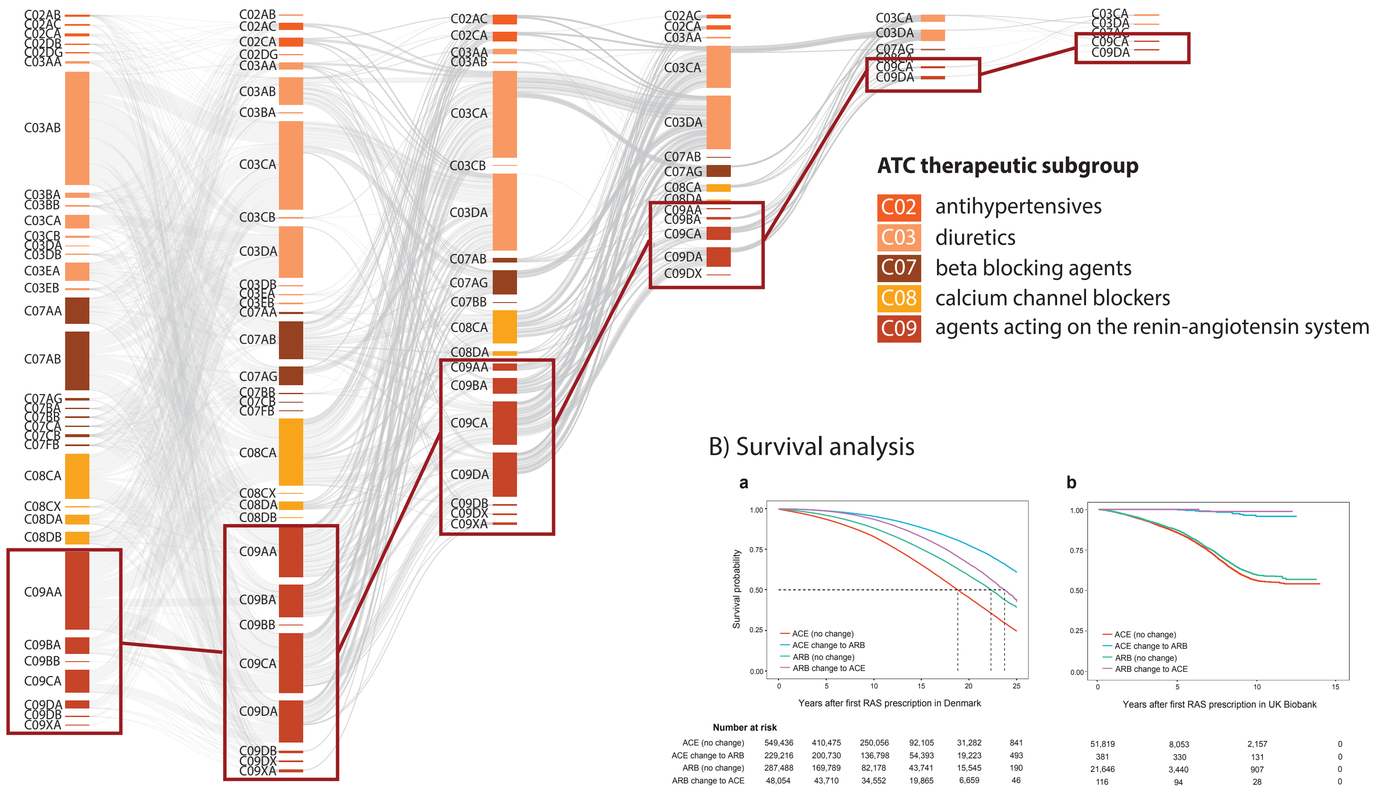


C) Construct longer trajectories

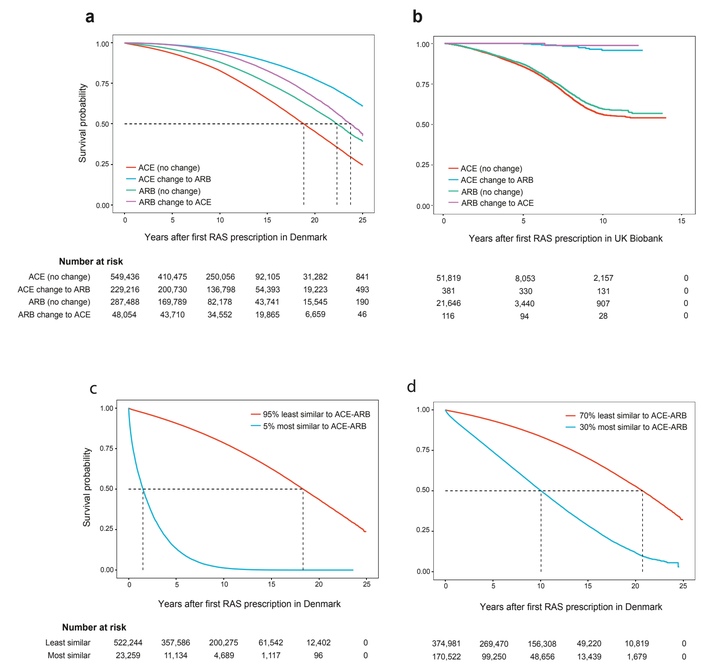


RESULTS

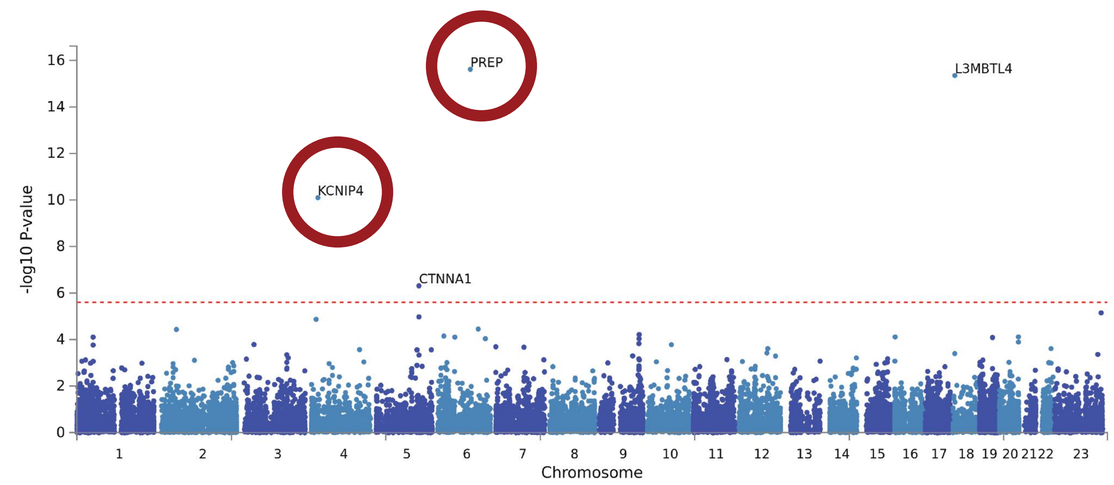
A) Hypertension trajectories



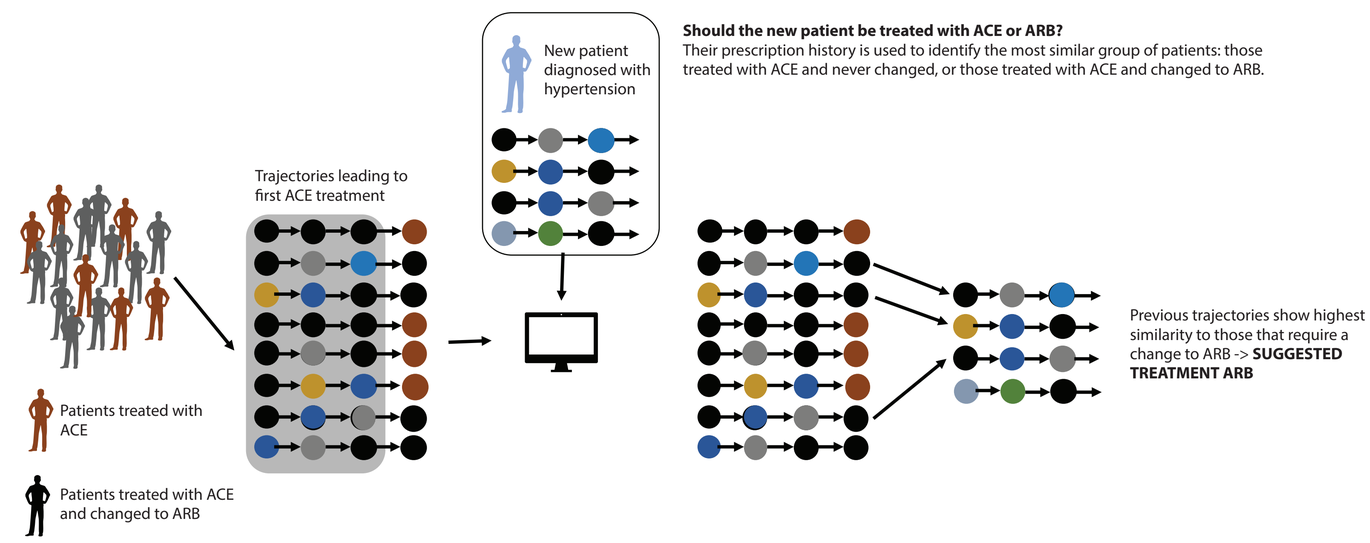
B) Survival analysis



C) Genetic differences discovered



D) Predictive model: Which drug should a patient take?



CONCLUSION

Drug pairs were used to build multi-step longitudinal prescription trajectories and subsequently used to stratify patients and calculate survival hazard ratio between the stratified groups. The similarity between prescription histories was used to determine individuals' best treatment option. Over the course of 122 million person years of observation, we identified 9 million common prescription trajectories and demonstrated their predictive power using hypertension as a case. Patients prescribed angiotensin converting enzyme (ACE) inhibitor without change, angotensin receptor blockers (ARBs) without change, ACE with posterior change to ARB, and ARB posteriorly changed to ACE. In adjusted time-to-event analysis, individuals treated with ACE compared to those treated with ARB had lower survival probability. Replication in UK Biobank data showed the same trends.

The Novo Nordisk Foundation Center for Protein Research
UNIVERSITY OF COPENHAGEN
Region Hovedstaden
STATISTICS DENMARK

Optimizing drug selection from a prescription trajectory of one patient.

Alejandro Aguayo-Orozco[1,2], Amalie Dahl Haue[1,3], Isabella Friis Jørgensen[1], David Westergaard[1,2,4], Pope Lloyd Moeley[1], Laust Hvas Mortensen[2,5]*, Søren Brunak[1]*
(1) University of Copenhagen, Novo Nordisk Foundation Center for Protein Research; (2) Statistics Denmark; (3)The Heart Center, Rigshopitalet - Copenhagen University Hospital; (4) Department of Obstetrics and Gynaecoogy, Amager Hvidorve Hospital, Copenhagen University Hospital.

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